

TPT Agent

Algorithm Documentation

Tradier Paper Trading — Full Strategy Reference

Strategy	Cash-Secured Puts (CSP) + LEAPS Deep ITM Calls
Broker	Tradier Sandbox (Paper Trading) — Account VA54665450
Schedule	Daily at 6:35 AM PT (every day)
Data	Tradier real-time APIs + yfinance (earnings only)
Discord	#beta-ai-trades — Summary cards only
Version	Generated June 14, 2026 (Updated with Phase 2 optimizations)

The TPT Agent is a fully automated options trading bot that runs on Tradier's paper trading sandbox. It executes two parallel options strategies on the same curated stock watchlist: Cash-Secured Puts (CSP Wheel strategy) and LEAPS deep ITM calls (stock replacement). The bot runs once per day at 6:35 AM PT, screens the entire approved stock universe, manages existing positions, and opens new ones based on a refined rules-based scoring system. **Phase 2 Optimizations (June 14, 2026):**

This version introduces **6 major enhancements** to improve capital efficiency, reduce API calls, and implement sophisticated position management: **#1 — Unified Stock Screening:** Single pass over all tickers replaces separate CSP/LEAPS passes. One quote + one history call per stock → ~50% API reduction.

#2 — RSI Hard Gate for CSP: $RSI \geq 65$ immediately disqualifies overbought stocks (no options lookup).

#3 — BB-Based Per-Stock Delta Selection: Delta range now driven by stock's Bollinger Band position (3 tiers: aggressive 0.25–0.35, moderate 0.15–0.25, conservative 0.08–0.15).

#4 — Scoring Refinement: Replaced price < \$100 criterion with $RSI < 50$ (actively oversold = better entry).

#5 — LEAPS Delta Target Lowered: Min 0.70, Max 0.85, Target 0.77 (from 0.80–0.99 targeting 0.85).

#6 — VIX Gate Refined: LEAPS enabled when $VIX \leq 18$ **OR** ≥ 21 (excludes 18–21 stress zone, added opportunity at high VIX).

Key Design Principles:

(1) Never trade below the 200-day SMA. (2) Never sell a CSP into an earnings window. (3) Size up only on high-conviction setups (score 5/5). (4) **New:** Close CSPs at 21 DTE with positive P&L; to avoid gamma. (5) **New:** Hard stop-loss at 2× premium received. (6) VIX gates both deployment amount and LEAPS eligibility. (7) **New:** Top 5 candidates only (CSP + LEAPS) to reduce noise.

Before opening any new positions, the bot scans every open position and closes those that have reached their exit triggers. **Phase 2 now implements 3 independent rules (any one triggers a close):**

Rule	Trigger (CSP)	Trigger (LEAPS)	Action
Rule #1 50% Capture	$(\text{Entry} - \text{Current}) / \text{Entry} \geq 50\%$	$(\text{Current} - \text{Entry}) / \text{Entry} \geq 25\%$	BUY/SELL TO CLOSE at live mid
Rule #2 21-DTE Exit	$\text{DTE} \leq 21$ AND $\text{P\&L} \geq -50\%$	$\text{DTE} \leq 200$ (N/A)	CLOSE to avoid gamma explosion
Rule #3 Stop-Loss	$\text{Current} \geq \text{Entry} \times 3.0$ (loss = $2 \times \text{Premium}$)	$\text{Premium} \geq 50\%$	Hard cap on losses

VIX → CSP Delta Range (Stock-Dependent in Phase 5):

CSP delta is **no longer global**. Instead, it's determined per-stock based on Bollinger Band position in Phase 5.

VIX Level	Deploy %	Rationale
>25	80%	Fear = opportunity, deploy aggressively
20–25	70%	Elevated vol, still good entry conditions
15–20	60%	Moderate vol, standard deployment
12–15	40%	Low vol, be selective and conservative
<12	20%	Very low vol, premiums too thin to deploy much

LEAPS VIX Gate (UPDATED): $VIX \leq 18$ OR $VIX \geq 21$

Why? Below 15 = market complacent, poor LEAPS entry. 15–18 = ideal (calm, stocks have pulled back). 18–21 = moderate stress, IV inflated unfairly. Above 21 = fear-driven opportunity (stocks deeply oversold, good LEAPS setup).

Budget Calculations:

- CSP Budget = Cash × VIX deploy %
- LEAPS Budget = Portfolio Value × 15% (raised from 10%) – Current LEAPS Exposure (never exceeds 15% of portfolio total)
- LEAPS Gate = Screening + execution ONLY when $VIX \leq 18$ OR $VIX \geq 21$

Key Change: All 54 tickers are now screened in ONE pass, not separately for CSP and LEAPS.

Process: Load approved tickers → For each ticker: fetch 310 days OHLCV + live quote → compute SMA20/50/200, Bollinger Bands, RSI, beta → apply hard filters (200 SMA + earnings) → store results in shared *stock_universe* dict.

CSP and LEAPS phases both read from this same dict, eliminating duplicate Tradier calls. **Result:** ~50% reduction in API calls (from ~216 to ~108 per run).

Step 1: RSI Hard Gate (NEW)

Before any scoring: If $RSI \geq 65$ → skip this stock entirely (overbought, high mean-reversion risk in 20–45 DTE window).

Step 2: BB-Based Delta Range (NEW — Replaces VIX-Global Range)

For this stock, determine CSP put delta range based on where price sits relative to Bollinger Bands:

BB Position	Delta Range	Rationale
Price $\leq$ Lower BB + 3%	0.25 – 0.35 (Aggressive)	Stock oversold → high reversal probability → safe to sell close to ATM
Between Lower & Mid BB	0.15 – 0.25 (Moderate)	Partial pullback → moderate entry quality → standard delta
Price $\geq$ Mid BB	0.08 – 0.15 (Conservative)	At/above average → more downside room → stay far CTM

Step 3: Technical Scoring (Max 5 Points)

- Above 50-day SMA → +1
- At/below 20-day SMA → +1
- Healthy pullback 0.5–5% today → +1
- **NEW:** $RSI < 50$ (actively oversold) → +1 (replaces "price < \$100")
- $IV \geq 40\%$ → +1

Minimum score to qualify: 3/5

Step 4: ARR Cap (UPDATED)

Max ARR: 70% (raised from 60%). If best ARR > 70%, step down to highest ARR $\leq$ 70%.

Step 5: Top 5 Selection (NEW)

Sort qualifying CSPs by: score DESC, then ARR DESC → Select top 5.

VIX Hard Gate (UPDATED): LEAPS screening runs **ONLY when $VIX \leq 18$ OR $VIX \geq 21$** . Excluded zone: $18 < VIX < 21$ (moderate stress, poor entry/exit conditions).

LEAPS Scoring (Unchanged — 3 Points)

- Earnings > 15 days away → +1
- Price near lower Bollinger Band (beta-adjusted) → +1
- RSI(14) < 70 → +1

Minimum score: 2/3

Contract Selection (UPDATED Delta Range)

- Expiration: 365–730 days (1–2 years)
- **NEW Delta Range:** 0.70 – 0.85 (was 0.80–0.99)
- **NEW Target Delta:** 0.77 (was 0.85)
- Pick contract closest to 0.77

Ranking & Top 5 Selection (NEW)

Sort by: leaps_score DESC, then RSI ASC (most oversold first) → Select top 5.

CSP Parameters:

Parameter	Default	Description
MIN_DTE	20	Minimum days to expiration
MAX_DTE	45	Maximum days to expiration
MIN_ARR	40%	Minimum annualized return on risk
MAX_ARR	70%	ARR cap (raised from 60%)
CSP_RSI_OVERBOUGHT	65	Hard gate — skip if RSI $\geq$ this
CSP_DTE_EXIT	21	Close if DTE $\leq$ this AND positive/neutral P&L
CSP_STOP_LOSS_MULT	2.0	Close if loss $\geq 2\times$ premium received
CSP_CLOSE_PREMIUM_PCT	50%	Close CSP when premium captured = this

LEAPS Parameters (UPDATED):

Parameter	Default	Description
LEAPS_MIN_DTE	365 days	Minimum DTE for LEAPS contracts
LEAPS_MAX_DTE	730 days	Maximum DTE for LEAPS (1–2 years)
LEAPS_MIN_DELTA	0.70	Minimum call delta (lowered from 0.80)
LEAPS_MAX_DELTA	0.85	Maximum call delta (lowered from 0.99)
LEAPS_TARGET_DELTA	0.77	Target delta (lowered from 0.85)
LEAPS_VIX_CALM_MAX	18	Upper bound for calm zone
LEAPS_VIX_FEAR_MIN	21	Lower bound for fear zone
LEAPS_MAX_PORTFOLIO_PCT	15%	LEAPS hard cap (raised from 10%)
LEAPS_CLOSE_PROFIT_PCT	25%	Close LEAPS at profit (raised from 10%)
LEAPS_STOP_LOSS_PCT	50%	Close LEAPS if down this much (NEW)

#1 — Unified Stock Screening

Single pass, shared data dict → ~50% API reduction

#2 — RSI Hard Gate for CSP

RSI $\geq$ 65 disqualifies overbought stocks immediately

#3 — BB-Based Per-Stock Delta Selection

3 tiers: Aggressive (0.25–0.35), Moderate (0.15–0.25), Conservative (0.08–0.15) based on BB position

#4 — Scoring Refinement

Replaced price < \$100 with RSI < 50 (actively oversold indicator)

#5 — LEAPS Delta Range Optimization

Min 0.70, Max 0.85, Target 0.77 (better leverage, safer OTM)

#6 — VIX Gate Expansion

VIX $\leq$ 18 **OR** $\geq$ 21 (captures high-vol opportunity, excludes stress zone)

#7 — LEAPS Budget Increase

15% of portfolio (from 10%)

#8 — LEAPS Profit Target Raised

+25% (from +10%)

#9 — 21-DTE Exit Rule

Close CSPs at DTE $\leq$ 21 with positive P&L; to avoid gamma explosion

#10 — 2× Premium Stop-Loss

Hard cap on CSP losses at 2× premium received

#11 — Top-5 Candidates Only

Limit to 5 CSP + 5 LEAPS per run to reduce noise and API usage

#12 — P&L; Calculation Fix

Corrected Tradier's negative cost_basis for short positions